

## MEFTAHUL JANNAT NUHA

Bachelor of Science (BSc) in Electrical and Electronic Engineering

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Links: [LinkedIn](#) | [ResearchGate](#) | [Google Scholar](#) | [ORCID](#)

### EDUCATION

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**Bachelor of Science (BSc) in Electrical and Electronic Engineering** | Rajshahi University of Engineering & Technology (RUET) | 2022 - Present

- Current **CGPA**: 3.41/4.00 (Till 7<sup>th</sup> Semester)

### THESIS WORK

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**Thesis Title:** Multimodal Sleep Stage Classification Using Deep Learning Models | June 2025 – Present

Role: Bachelor's Thesis Student

Supervisor: Dr. Mohammad Abdul Motin

Institution: Rajshahi University of Engineering & Technology, Rajshahi-6204, Bangladesh

**Thesis Overview:** Developing a deep learning-based methodology for automatic sleep stage classification using multimodal physiological signals by processing and analyzing PPG, EDA, and accelerometer data. The framework is designed for resource-efficient deployment in wearable and edge devices and is suitable for highly imbalanced datasets commonly found in physiological signal analysis.

### RESEARCH INTERESTS

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- Biomedical Signal Processing
- Wearable Health Monitoring Systems
- Machine Learning application for Healthcare
- Embedded AI and Edge Computing
- Internet of Medical Things (IoMT)

### RESEARCH EXPERIENCES & PUBLICATIONS

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- **IEEE Sensors Journal (2026)**  
**Title:** *TinyArtifactNet: A Resource-Efficient Deep Learning Model for On-Device Motion Artifact Detection in PPG*  
**Role:** Methodology, Deep learning Model building, Evaluation and Manuscript Preparation  
- Proposed a lightweight deep learning model for real-time motion artifact detection in PPG signal on wearable devices.  
**Publication Date:** May, 2026  
DOI: [10.1109/JSEN.2026.3692462](https://doi.org/10.1109/JSEN.2026.3692462)
- **Data in Brief (2025)**  
**Title:** *A Multi-modal Wearable Dataset for Cognitive Attention and Task-based Stress Analysis*  
**Role:** Data curation and original draft writing  
- Developed and documented a multimodal physiological dataset for cognitive attention and stress research.  
**Publication Date:** December 2025  
DOI: [10.1016/j.dib.2025.112295](https://doi.org/10.1016/j.dib.2025.112295)

## PROFESSIONAL EXPERIENCES & LEADERSHIP

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**Research Assistant** | RUET SPML Lab, Rajshahi University of Engineering & Technology (RUET) | August 2024 – Present

- Conducting research on biomedical signal processing and machine learning on edge IOT devices

**Head of Shutdown and Safety** | Team Crack Platoon (Formula SAE Team) | March 2026 – June 2026

- Leading safety protocols and shutdown procedures for formula student car

**Electrical Safety Officer (ESO)** | FSAE Japan 2025

- Earned ESO certification for ensuring electrical safety during competition

**Chief of Low Voltage System** | Team Crack Platoon | September 2024 – March 2026

- Managed electrical and battery management systems for Formula SAE vehicle

**Class Representative** | Electrical & Electronic Engineering | September 2024 – August 2025

- Served as liaison between faculty and students, addressing academic and departmental concerns

## TECHNICAL COMPETITIONS

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- IEEE Signal Processing Cup: “Deepfake Face Detection In The Wild” as a member of Team “Pseudo Spectrum” (Team ID: 28267)
- Formula SAE Japan 2025: Participated in EV segment (EV026) as a Team member and Electrical Safety Officer (ESO)

## SKILLS

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- **Programming & Development:** Python, MATLAB, C/C++
- **Machine Learning & AI:** Deep Learning, TensorFlow / Keras, CNN, LSTM, Transformer Model
- **Research & Analysis:** Experimental design, Scientific Writing, Data Analysis, Statistical Analysis
- **Biomedical Signal Processing:** PPG Signal Processing, Feature Extraction & Data Preprocessing
- **Tools & Software:** Git & GitHub, Jupyter Notebook, Kaggle, MATLAB
- **Embedded Systems:** STM32 development, real-time signal processing, sensor interfacing
- **Hardware & Electronics:** Circuit Design, Sensor Integration, Electrical Safety Systems
- **Leadership & Management:** Team coordination, Research collaboration, Project planning

## TRAINING EXPERIENCES

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- Industrial trainee at Civil Aviation Authority of Bangladesh (CAAB)
- Industrial trainee at THiNK, Dhaka, Bangladesh

## PROJECTS & WORKSHOPS

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- Electronic Shop Practice event (EEE-3100): IoT based Wireless solar charging station for electric vehicle (Prototype).
- Capstone Project I (EEE-4100): TinyMA-Net: Lightweight deep learning model for real-time motion artifact detection in PPG signals on embedded systems

## CERTIFICATIONS

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- Electrical Safety Officer (ESO) – FSAE Japan 2025

- Class Representative – Recognition certificate for representation and service, Rajshahi University of Engineering & Technology (RUET)

## LANGUAGE SKILLS

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- English
- Bangla